

CROSSING TIMES IN CUMULATIVE PROCESSES

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Summary

In this paper cumulative processes $X(t) = \sum_{n=1}^{N(t)} Y_n$ are considered in which the terms Y_n are gamma distributed and the number $N(t)$ of terms in the sum up to time t follows a compound Poisson process. The probability density function (p.d.f.) and the moments of $T(x)$, the time needed by the process $X(t)$ to cross a given level x , are derived. In particular, it is shown that for constant, respectively gamma distributed intensity of the associated Poisson process the p.d.f. of $T(x)$ is the countable sum of gamma, respectively compound gamma p.d.f.'s each weighted by a term depending on the distribution of Y_n only.

1. Let $N(t)$, $[0, \infty)$ be a counting process and let $Y_n, n \in \mathbb{N}$ be a sequence of independently identically distributed random variables. Assume that $N(t)$ and $\{Y_n : n \in \mathbb{N}\}$ are independent. The process $X(t)$, $[0, \infty)$ defined by

$$X(t) := \sum_{n=1}^{N(t)} Y_n$$

is then called the cumulative process associated with the counting process $N(t)$, $[0, \infty)$. For a physical interpretation of a cumulative process we refer to the examples given by Parzen (1962), p.129-130, and by Cox (1962), p. 91-92. Recently Uhler and Bradley (1970) applied a model for petroleum exploration in which it is hypothesised that the spatial occurrence of petroleum reservoirs is represented by a non-homogeneous Poisson process and that the sizes of individual reservoirs are log-normally distributed. In this case $X(t)$ represents the total petroleum reserves in a space unit of size t .

In most applications, however, t will stand for time, in which case $T(x)$, the time taken by the process $X(t)$ to cross a certain level x , is of particular interest. (Note that a recrossing of a level is impossible because the Y_n are assumed to be positive random variables). As shown in Steinijans (1970) the processes $X(t)$ and $T(x)$ are inverse processes of each other, which means that the sample functions of the process $T(x)$ are the inverse functions of the sample functions of the process $X(t)$, and vice versa.

2. Using Laplace transforms Cox (1962), p.97-99, has shown that in the case where $N(t)$ is a Poisson process with intensity $\nu > 0$ and the Y_n are exponentially distributed (with parameter $\lambda > 0$), the p.d.f. of $T(x)$ is given by

$$(1) \quad f_{T(x)}(t) = \nu e^{-\nu t - \lambda x} I_0(2\sqrt{\nu t \lambda x}),$$

where for real x and $s \geq -1$

$$I_s(x) = \sum_{n=0}^{\infty} \frac{1}{n! \Gamma(n+s+1)} \left(\frac{x}{2}\right)^{2n+s}$$

denotes the Bessel function of pure imaginary argument.

We shall generalise (1) as follows. Firstly we will allow a random mixture of events of Poisson type, that is we consider a non-homogeneous Poisson process for which the intensity ν is a positive random variable with distribution function $F_\nu(\nu)$. Secondly, we will assume that the Y_n have a gamma distribution with parameters $\ell \in \mathbb{N}$ and $\lambda > 0$. In this case the conditional distribution of $X(t)$ given $N(t) = n$ is gamma with parameters $n\ell$ and λ , hence

$$f_{X(t)}(x) = \begin{cases} \sum_{n=1}^{\infty} \lambda \frac{(\lambda x)^{n\ell-1}}{(n\ell-1)!} e^{-\lambda x} \int_0^{\infty} e^{-\nu t} \frac{(\nu t)^n}{n!} dF_\nu(\nu), & x > 0 \\ \int_0^{\infty} e^{-\nu t} dF_\nu(\nu) & , x = 0 \end{cases}$$

From the relation

$$T(x) \leq t \text{ iff } X(t) \geq x$$

one obtains by

$$\int_x^{\infty} e^{-\lambda u} \frac{(\lambda u)^{n\ell-1}}{(n\ell-1)!} d\lambda u = e^{-\lambda x} \sum_{i=0}^{n\ell-1} \frac{(\lambda x)^i}{i!}$$

$$F_{T(x)}(t) = \sum_{n=1}^{\infty} \sum_{i=0}^{n\ell-1} e^{-\lambda x} \frac{(\lambda x)^i}{i!} \int_0^{\infty} e^{-\nu t} \frac{(\nu t)^n}{n!} dF_\nu(\nu), \quad x > 0.$$

Finally, differentiation with respect to t and simple manipulation yield

$$(2) \quad f_{T(x)}(t) = \sum_{n=0}^{\infty} w_n^{(\ell)} \int_0^{\infty} \nu e^{-\nu t} \frac{(\nu t)^n}{n!} dF_\nu(\nu), \quad x > 0,$$

where

$$w_n^{(\ell)} = \sum_{i=0}^{\ell-1} e^{-\lambda x} \frac{(\lambda x)^{n\ell+i}}{(n\ell+i)!}$$

If $\ell=1$ and if the associated counting process is a homogeneous Poisson process, i.e. for constant ν , (2) reduces to (1) which can also be written as

$$f_{T(x)}(t) = \sum_{n=0}^{\infty} w_n^{(1)} f_{n+1, \nu}(t),$$

where $f_{n+1, \nu}(t)$ denotes the p.d.f. of a gamma distribution with parameters $n+1$ and ν . (Note that $f_{T(x)}(t)$ is the countable sum of gamma p.d.f.'s each weighted by a term from a Poisson distribution).

If ν itself is gamma distributed with p.d.f.

$$(3) \quad f_{\nu}(\nu) = \rho \frac{(\rho \nu)^{m-1}}{\Gamma(m)} e^{-\rho \nu}; \quad \rho, m > 0; \nu > 0,$$

then (2) reduces to

$$f_{T(x)}(t) = \sum_{n=0}^{\infty} w_n^{(\ell)} f_{n+1, m, \rho}(t),$$

where

$$f_{a, \gamma, \delta}(t) = \frac{\delta \gamma t^{a-1}}{B(a, \gamma)(\delta + t)^{a + \gamma}}; \quad a, \gamma, \delta > 0$$

denotes the compound gamma distribution as introduced by Dubey (1970). (Note that for $\ell=1$ $f_{T(x)}(t)$ becomes the countable sum of compound gamma p.d.f's each weighted by a term from a Poisson distribution).

In this context it should be mentioned that Uhler and Bradley used the parameter distribution (3) in order to account for the tendency of petroleum reservoirs to cluster.

3. Provided

$$\mu'_{-k} := \int_0^{\infty} \nu^{-k} dF_{\nu}(\nu)$$

exists, the k -th moment of $T(x)$ is given by

$$(4) \quad \mu'_k(x) = E[T(x)^k] = \mu'_{-k} \sum_{n=0}^{\infty} w_n^{(\ell)} (n+1)_k, \quad k=1, 2, \dots,$$

where

$$(a)_0 = 1; \quad (a)_k = a(a+1) \cdots (a+k-1), \quad k \geq 1.$$

For constant ν one has

$$\mu'_{-k} = \nu^{-k},$$

and for the parameter distribution (3)

$$(5) \quad \mu'_{-k} = \frac{\rho^k}{(m-k)_k}, \quad m > k.$$

In particular, for $\ell=1$ and constant ν

$$E[T(x)] = \frac{1}{\nu}(1 + \lambda x), \quad V[T(x)] = \frac{1}{\nu^2}(1 + 2\lambda x),$$

and the moment generating function of $T(x)$ is given by

$$m_{T(x)}(u) = \frac{\nu}{\nu - u} e^{\lambda x \frac{u}{\nu - u}}, \quad |u| < \nu.$$

For the parameter distribution (3) one obtains in the case $\ell=1$

$$E[T(x)] = \frac{\rho}{m-1} (1 + \lambda x), \quad m > 1,$$

$$V[T(x)] = \frac{\rho^2}{(m-1)^2(m-2)} (m + 2m\lambda x + (\lambda x)^2), \quad m > 2,$$

and it can be seen from (5) together with (4) that the moment generating function does not exist in this case.

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